

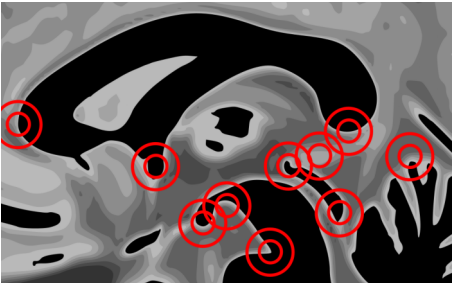
01

THE DATA

protocol · consensus ground truth · reference frames

Taha et al., 2023

THE PROTOCOL



32 landmarks · 8 regions  
AFIDs protocol · Lau et al., 2019

IMAGES & TEMPLATES

132

subject images

+ 15

human templates

+ 6

macaque templates

4,896

curated fiducials

153 brains × 32 landmarks · 4 open datasets

CONSENSUS = GROUND TRUTH



per-subject consensus  
from 492 expert annotations

RATER-TO-CONSENSUS ERROR, PER LANDMARK

AC

SPLE

PMJ

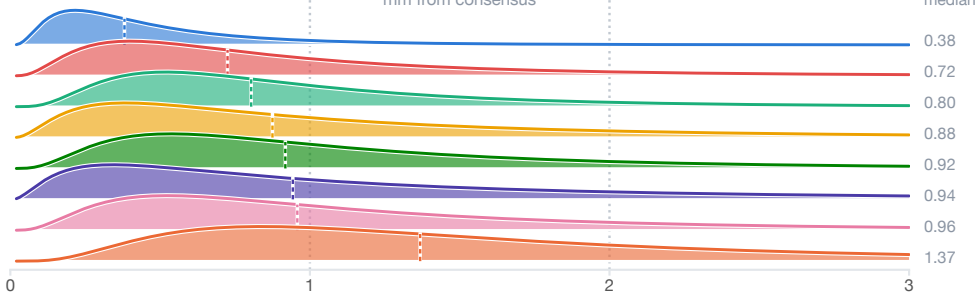
CUL

PG

RLVAC

RSAMTH

LIGO



mm from consensus

median

0.38  
0.72  
0.80  
0.88  
0.92  
0.94  
0.96  
1.37

median 0.87 mm

57% < 1 mm

82% < 2 mm

difficulty is landmark-specific — tutor and validator both read against it

One Euclidean error — two uses

the same millimetre, always read against how hard the landmark is

taught in context

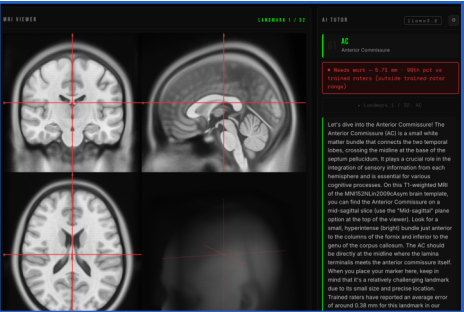
audited at scale

02

LEARN

AI-guided tutor · /learn

bring your own API key



NiiVue viewer + streaming AI tutor

1

2

3

4

5

Introduce → Place → Compute → Feedback → Dialogue

repeat × 32 landmarks

anatomy-first

viewer-aware

RAG-grounded

productive failure

"Your AC sits 1.2 mm from consensus — the 94th percentile, where trained raters land within 0.38 mm.  
Re-check the midline: AC is where the commissure crosses it, not the bright bundle above."

— streaming /learn tutor

03

VALIDATE

multi-template engine

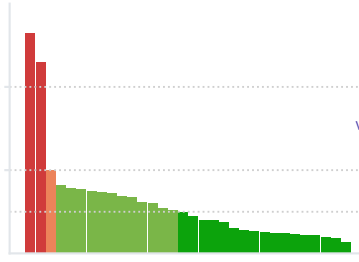
instant feedback

upload

FCSV / JSON

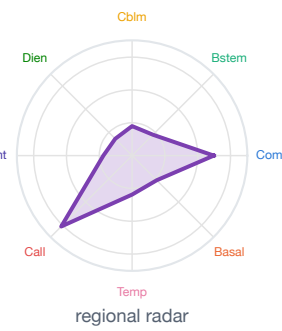
32 errors

vs 21 templates



ranked error (mm)

< 1 excellent    1-2 good    2-4 fair    ≥ 4 needs work



regional radar

< 1 excellent    1-2 good    2-4 fair    ≥ 4 needs work

CALIBRATED REPORT

| landmark | error  | verdict           |
|----------|--------|-------------------|
| LIGO     | 1.2 mm | excellent · 43rd  |
| AC       | 1.2 mm | off-target · 94th |
| PMJ      | 0.9 mm | good · 57th       |
| SPLE     | 3.4 mm | off-target · 98th |

OPEN BY DESIGN

GPL-3.0 code

Open reference data

TemplateFlow

NiiVue viewer

3D Slicer

ORCID identity

Docker self-host